

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\in \setminus (Q^\circ \in)) : EA \rightarrow EA$ (8.1).

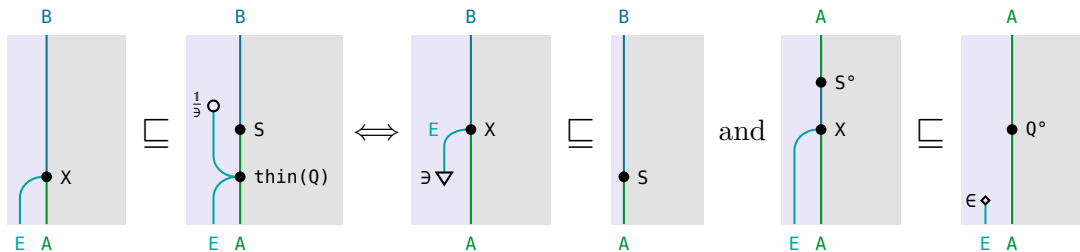
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: $\in X$ runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \in$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

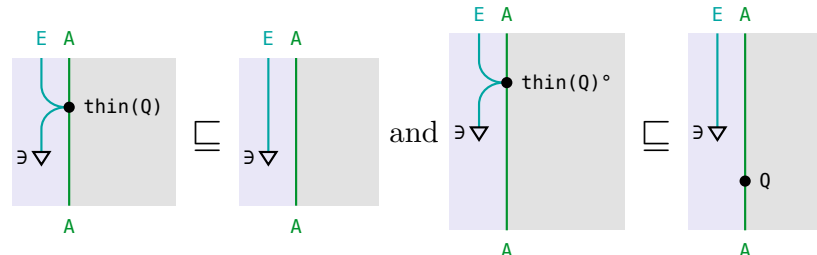
the law	what it says
$X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \sqsubseteq Q^\circ \in$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \in \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \in \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$ $QQ \sqsubseteq Q \implies \text{thin}(Q) \text{thin}(Q) \sqsubseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge RR \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \sqsubseteq R, 1 \sqsubseteq Q, RR \sqsubseteq R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \sqsubseteq R$ follows	thin-introduction: thinning first cannot lose an R -minimum
$\text{thin}(Q) \sqsupseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination: keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{union} \sqsubseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \sqsubseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \sqsubseteq Q^\circ \in$

(8.1.1a)



(8.1.1b)



(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$, since $\exists \exists \exists = \Lambda(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \sqsubseteq R, 1 \sqsubseteq Q, RR \sqsubseteq R$

(8.1.2a)

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a)

(a)	$\text{thin}(Q) \text{ est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — (8.1a)}$
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(8.1.2b)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) (b)

(a)	$\text{thin}(Q) \exists$	$\text{est}(R) \sqsubseteq \exists \text{ — est-laws}$
(b)	$\exists$	$\text{thin}(Q) \exists \sqsubseteq \exists \text{ — (8.1.1a)}$

(8.1.2c)

$\sqsubseteq \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$
the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) (b) (c)

(a)	$Q^\circ \sqsubseteq \text{est}(R)$	$\sqsubseteq \text{thin}(Q) \sqsubseteq Q^\circ \sqsubseteq \text{ — (8.1.1a)}$
(b)	$Q^\circ R^\circ$	$\sqsubseteq \text{est}(R) \sqsubseteq R^\circ \text{ — est-laws}$
(c)	R°	$R \text{ transitive}$

(8.2)

$\text{est}(Q) \frac{1}{3} \sqsubseteq \text{thin}(Q)$
the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{3} \sqsubseteq \exists$
every member of the singleton is a member of the set

(a) (b)

(a)	$\text{est}(Q) \frac{1}{3} \sqsubseteq \exists$	$\triangleq \text{thin}, \exists/\exists \text{ half}$
(b)	$\exists$	$\frac{1}{3} \sqsubseteq 1 \text{ — pow-laws; } \text{est}(Q) \sqsubseteq \exists \text{ — est-laws}$

$\sqsubseteq \text{est}(Q) \frac{1}{3} \sqsubseteq Q^\circ \sqsubseteq$
the singleton's member stands in Q to every member of the set

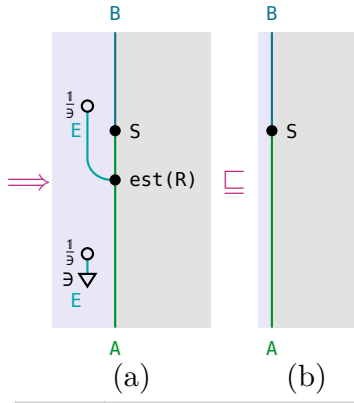
(a) (b) (c)

(a)	$\sqsubseteq \text{est}(Q) \frac{1}{3}$	$\triangleq \text{thin}, \sqsubseteq \backslash (Q^\circ \sqsubseteq) \text{ half}$
(b)	$Q^\circ \frac{1}{3}$	$\sqsubseteq \text{est}(Q) \sqsubseteq Q^\circ \text{ — est-up}$
(c)	$Q^\circ \sqsubseteq$	$\frac{1}{3} \sqsubseteq \sqsubseteq \text{ — pow-laws}$

$$\mathbf{RnS^\circ S \sqsubseteq Q \implies \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q)} \quad (8.3)$$

given $\mathbf{Rn(S^\circ S) \sqsubseteq Q}$, Q a preorder

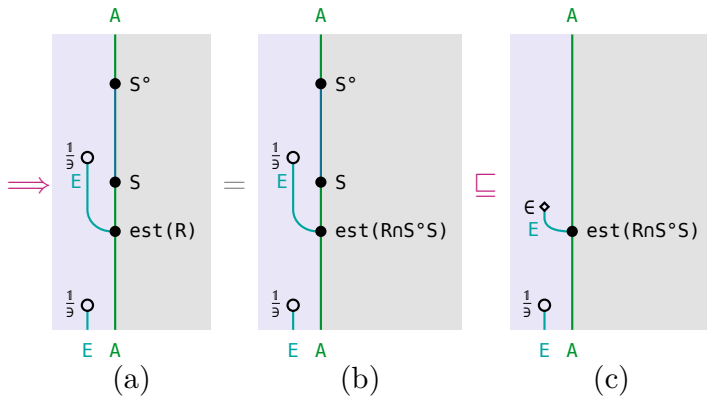
$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \exists S$
every member of the singleton is a value S returns



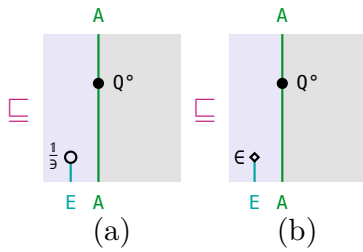
(a)	$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$	(8.1a), first condition
(b)	S	$\frac{1}{S} \ni \mathbb{1}$ — pow-laws; $\frac{S}{S} \text{est}(R) \sqsubseteq S$ — est-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \implies S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \epsilon}$$

the singleton's member stands in Q to every value S returns at the same argument



(a)	$S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$	(8.1a), second condition
(b)	$S^\circ \frac{S}{S} \text{est}(RnS^\circ S) \frac{1}{S}$	$\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(RnS^\circ S)$ — est-laws
(c)	$\epsilon \text{est}(RnS^\circ S) \frac{1}{S}$	$S^\circ \frac{S}{S} \sqsubseteq \epsilon$ — pow-laws



(a)	$Q^\circ \frac{1}{S}$	$\epsilon \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$ — est-up; $Rn(S^\circ S) \sqsubseteq Q$
(b)	$Q^\circ \epsilon$	$\frac{1}{S} \sqsubseteq \epsilon$ — pow-laws

$Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$ Theorem 8.1		
thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder		
$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$ and $\langle \frac{(S)}{\exists} \rangle \circ \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \in$ (8.1a) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle \frac{(S)}{\exists} \rangle$ for its S		
(a)	$S^\circ F(Q^\circ \in \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle)$	fusion, hylomorphism theorem
(b)	$Q^\circ S^\circ F(\in \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle)$	$S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — mon-str, relator-laws
(c)	$Q^\circ \in \text{thin}(Q)$	$S^\circ F(\in) \langle \frac{F(\exists)S}{\exists} \rangle \sqsubseteq \in$ — pow-laws
(d)	$Q^\circ Q^\circ \in$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — $\triangleq \text{thin}$, adj-all

$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$ Corollary 8.1		
the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders		
(a)	$\langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	thinning theorem
(b)	$\langle \frac{(S)}{\exists} \text{est}(R) \rangle$	$\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — (8.1a)

§8.2. Paths in a layered network

(8.2a)

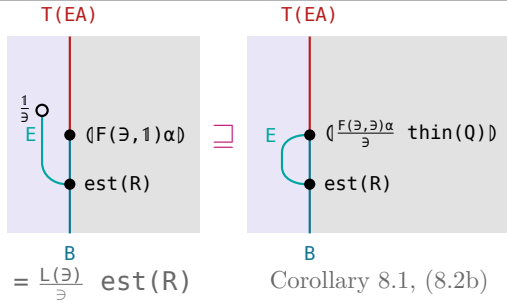
$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.
 $\text{wrap} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{cons} w(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.
 $\text{cost} \triangleq \langle [\text{wrap} z, \text{cons} w] \rangle \pi_2$, $\langle [\text{wrap} z, \text{cons} w] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.
 $Q \triangleq R n(\text{head head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$.

(8.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R n(S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \in R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \subseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \quad (8.2c)$$

a least-cost path in a layered network, as a fold over the layers



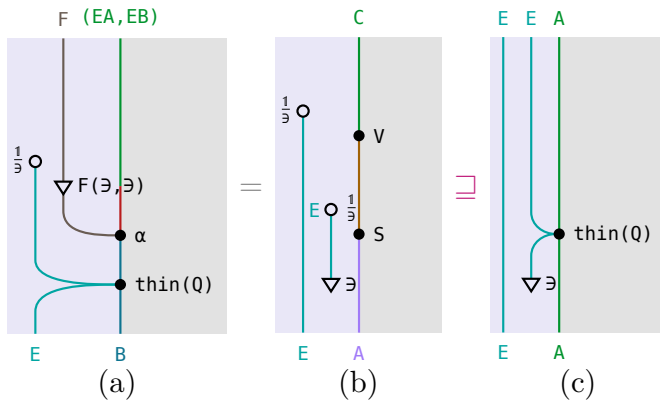
$\sqsubseteq$ (8.2d) under $\downarrow$ $\downarrow$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \downarrow \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \downarrow \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

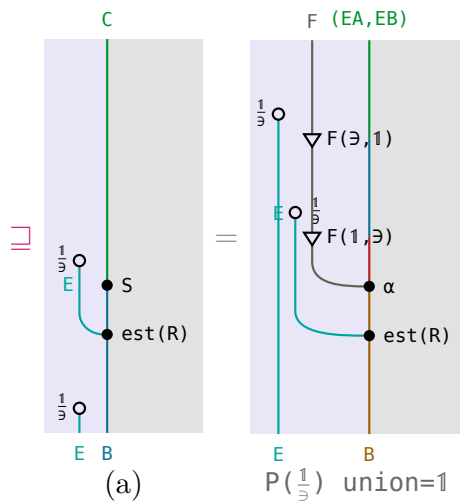
thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



(a)	$\frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$	the algebra of (8.2c) row 2
(b)	$\frac{V}{\exists} P(\frac{S}{\exists}) \text{union}$	$\frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}$
(c)	$P(\text{thin}(Q)) \text{union}$	$\text{union thin}(Q) \sqsupseteq P(\text{thin}(Q)) \text{union} \text{---} (8.1a)$

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

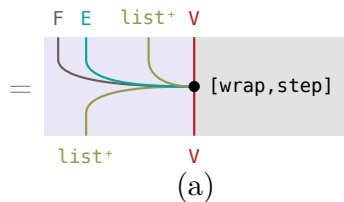
choosing an R-least member of each of those sets is below thinning each and taking the union



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \text{---} (8.1a), (8.2b)$
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$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



(a)	$[\text{wrap}, \text{step}]$	$\alpha \text{ a map; } \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \text{---} \triangle \text{ cost}$
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$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{---} \triangle \text{ cost}$$

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\llbracket \text{nil}, \text{bump } Q \rrbracket$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

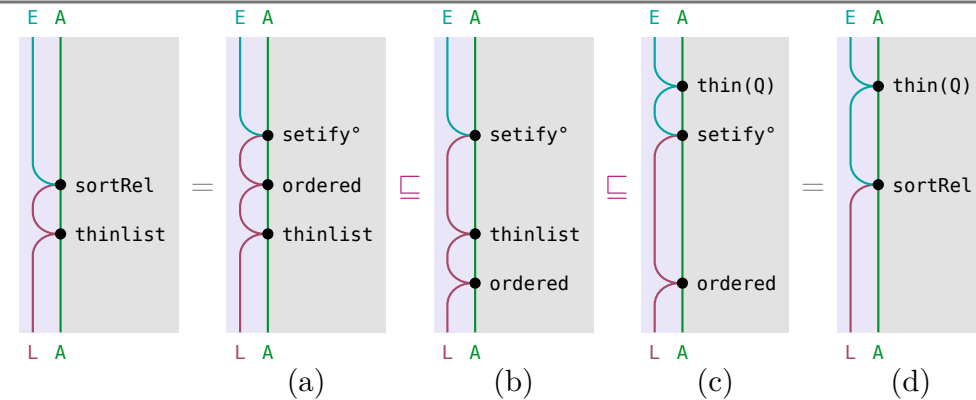
(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ } xs = [\text{minlist}(Q) \text{ } xs]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered subseq $\sqsubseteq$ subseq ordered $\wedge$ thinlist (8.6)

setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sortRel thinlist $\sqsubseteq$ thin(Q) sortRel

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.

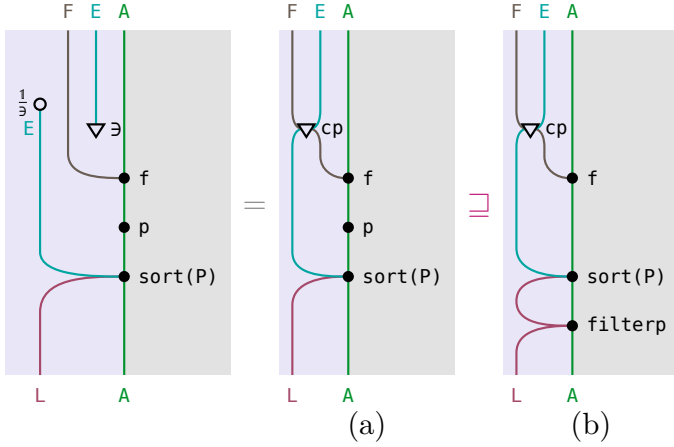


(a)	$\text{setify}^\circ \text{ ordered thinlist}$	$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
(b)	$\text{setify}^\circ \text{ thinlist ordered}$	$\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
(c)	$\text{thin}(Q) \text{ setify}^\circ \text{ ordered}$	$\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist, dom-laws, triple-chains}$
(d)	$\text{thin}(Q) \text{ sortRel}$	$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

Lemma 8.1

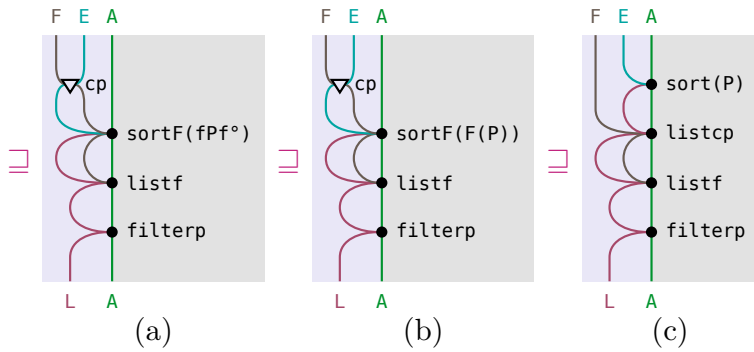
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge$
 $\text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$
 $\text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic
 on P , p coreflexive, F linear.

$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp } P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



(a)	$\text{cp } P(f) E(p) \text{sort}(P)$	$\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp) \text{ — pow-laws, } \triangleq \text{thinlist}$ $E(f) = P(f), f \text{ a map — powrel-laws}$
(b)	$\text{cp } P(f) \text{sort}(P) \text{filterp}$	$\text{sort}(P) \text{filter}(p) \sqsubseteq E(p) \text{sort}(P) \text{ — (8.3b)}$

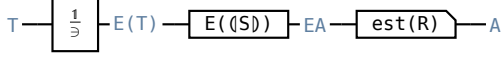
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$
 $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp } \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp } P(f) \text{sort}(P) \text{filterp}$
 combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



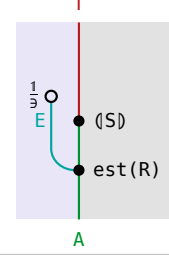
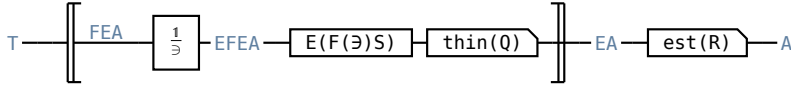
(a)	$\text{cp } \text{sortF}(fP f^\circ) \text{listf} \text{filterp}$	$\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P) \text{ — (8.3b)}$
(b)	$\text{cp } \text{sortF}(F(P)) \text{listf} \text{filterp}$	$FP \sqsubseteq fP f^\circ \text{ — mon-str, } \triangleq \text{thinlist}$
(c)	$F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp}$	$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP) \text{ — (8.3b)}$

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \subseteq R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge \text{MonoAlg}(f_2 \text{ } p_2, Q)$ Theorem 8.2
 $\wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1$
 $P f_1^\circ) \text{listf}_1 \subseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 P f_2^\circ) \text{listf}_2 \subseteq P(f_2) \text{sort}(P) \wedge$
 $\text{sort}(P) \text{filterp}_1 \subseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \subseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \subseteq \text{cp}$
 $\text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup } \text{sort}(P) \wedge$
 $\text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \subseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filterp}_1 \wedge$
 $g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp}(g_1, g_2) \text{merge}(P) \text{thinlist}(Q))$
 $\text{minlist}(R) \subseteq \frac{(S)}{3} \text{ est}(R)$
 a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

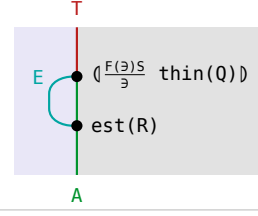
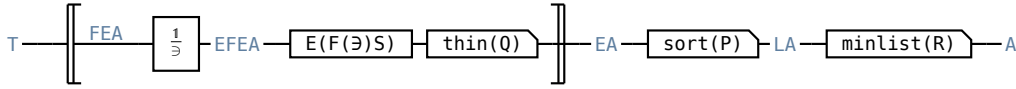
circuit



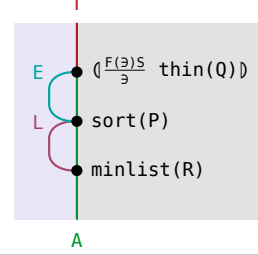
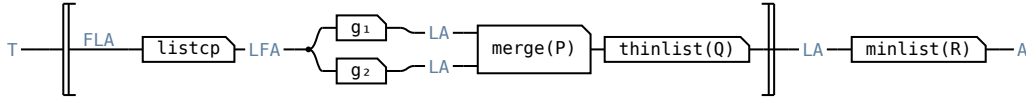
Hinze–Marsden


 $\sqsubseteq$


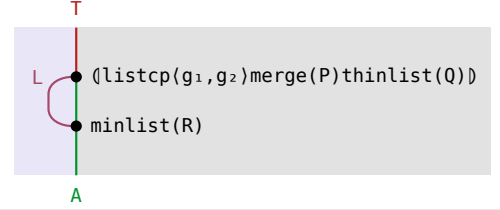
Corollary 8.1 at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$


 $\sqsubseteq$


$\text{sort}(P) \text{ minlist}(R) \subseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$


 $\sqsubseteq$


fusion at (8.3c)

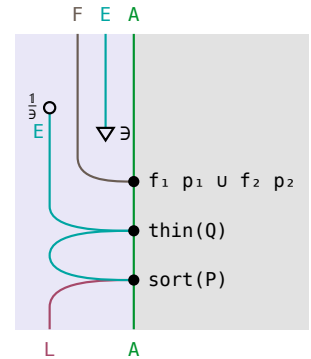
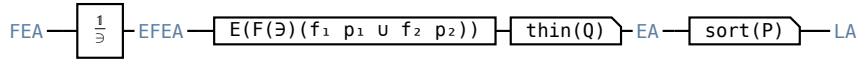


$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(\text{F}(P)) \\
& \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \\
& \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \sqcup f_2 \text{ p}_2)}{\exists} \\
& \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

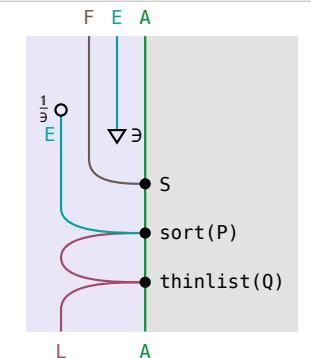
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)S)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \text{ --- (8.3b)}
\end{aligned}$$



$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{cup}} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& = \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle \text{cup} \text{sort}(P) \text{ thinlist}(Q) \\
& S = (f_1 \text{ p}_1) \sqcup (f_2 \text{ p}_2) \text{ --- } \triangleq \text{thinlist, then } \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{ thinlist}(Q) \\
& (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ --- (8.3b)}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists} \text{sort}(P), \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \text{sort}(P) \right\rangle \text{merge}(P) \text{ thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{ sort}(P), Y \text{ sort}(P) \rangle \text{ --- bdm-prod-laws}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\text{F}(\text{sort}(P))} \text{---} \text{FLA} \text{---} \boxed{\text{listcp}} \text{---} \text{LFA} \text{---} \boxed{\text{listf}_1} \text{---} \boxed{\text{filterp}_1} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\text{F}(\text{sort}(P))} \text{---} \text{FLA} \text{---} \boxed{\text{listcp}} \text{---} \text{LFA} \text{---} \boxed{\text{listf}_2} \text{---} \boxed{\text{filterp}_2} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{F}(\text{sort}(P)) \text{ listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ --- bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \\
& \text{filter}(p_i) \text{ --- } \triangleq \text{thinlist}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

`vol, wt : Item → Real, value ≐ list(vol) sum, weight ≐ list(wt) sum .`

`subseq = ([nil, cons] ∪ [nil, π2]) ∘ , ,` within `w` the coreflexive on `xs` with weight `xs ≤ w`, `0 ≤ w`.

`R ≐ value ≥ value°`, `Q ≐ Rn(weight ≤ weight°)`, `P ≐ R`, .

`FA = 1 + Item × A`, `listcp = wrap + cpr`, `g1 ≐ list([nil, cons])` `filter(within w)` `= [list(nil), h1]`,
`g2 ≐ list([nil, π2]) = [list(nil), h2]`.

`h1 ≐ list(cons) filter(within w)`, `h2 ≐ list(π2)`.

(8.4b)

the law	what it says
$(1 \times R) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w)R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w)Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned} & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \subseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\ & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \subseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP} \text{filterp}_1 \subseteq E(\text{within}) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \subseteq \text{cp} \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \subseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \subseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \subseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{mergeP} \\ & \text{thinlist}) \text{minlist} \subseteq \frac{\text{subseq} \text{ within}}{\exists} \text{est}(R) \end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
$ \begin{array}{c} \boxed{\text{[Item]}} \text{---} \boxed{\frac{1}{\exists}} \text{---} \boxed{E(\text{subseq within})} \text{---} \boxed{E[\text{Item}]} \text{---} \boxed{\text{est}(R)} \text{---} \boxed{\text{[Item]}} \\ \text{subseq (within w)} \\ \exists \quad \text{est}(R) \end{array} $	
$ \begin{array}{c} \boxed{\text{[Item]}} \text{---} \boxed{\frac{1}{\exists}} \text{---} \boxed{E(([\text{nil}, \text{cons}] \text{within } u \text{ } [\text{nil}, \pi_2]) \text{)}} \text{---} \boxed{E[\text{Item}]} \text{---} \boxed{\text{est}(R)} \text{---} \boxed{\text{[Item]}} \\ = \quad \frac{([\text{nil}, \text{cons}] \text{within } w) \cup [\text{nil}, \pi_2]}{\exists} \quad \text{est}(R) \\ \text{fusion, weights non-negative.} \end{array} $	
$ \begin{array}{c} \boxed{\text{[Item]}} \text{---} \boxed{\frac{\text{Item}}{\text{list}[\text{Item}]}} \text{---} \boxed{\text{listcp}} \text{---} \boxed{\text{list}(F[\text{Item}])} \text{---} \boxed{\text{listf}_1} \text{---} \boxed{\text{filterp}_1} \text{---} \boxed{\text{list}[\text{Item}]} \text{---} \boxed{\text{mergeP}} \text{---} \boxed{\text{thinlist}} \text{---} \boxed{\text{list}[\text{Item}]} \text{---} \boxed{\text{minlist}} \text{---} \boxed{\text{[Item]}} \\ \sqsupseteq \quad \text{Q(listcp } \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist}(Q)) \text{ minlist}(R) \\ \text{binary thinning theorem, at } P \triangle R, F \text{ linear, } Q \text{ from (8.4b)} \end{array} $	
$ \begin{array}{c} \text{Q}([\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)]) \text{ minlist}(R) \\ = \quad \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ --- } \triangle \text{ within; minlist}(R) \\ \text{is head, packings coming out in descending value} \end{array} $	

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$ $\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ, .$	both halves are monotonic on Q once ties in waste are broken by the first line
merge $T = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(8.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$
 $\text{wrap,new}]T[\text{wrap wrap,new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap,new}]) \text{sortP} \wedge \text{sortF}([\text{wrap wrap,glue}]T[\text{wrap}$
 $\text{wrap,glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap,glue}]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cpr}$
 $\text{sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len},w)) \text{sortP} \wedge \text{sortP}$
 $\text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{minlist} \sqsubseteq \frac{\text{partition fits}}{3} \text{est}(R))$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{3} \text{est}(R)$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue} ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u (\text{glue } (ok \ w))])}{3} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue} ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u ([\text{wrap wrap,glue} (ok \ w))])}{3} \text{est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}$.	
$[\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)] \text{minlist}(R)$ binary thinning theorem, at $P \triangleq T$ with merge $T = \text{cat}$, Q from (8.5b)	
$[\text{start, cpr } (h_1, h_2) \text{ cat thinlist}(Q)]$ $= \text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct — $\triangleq \text{partition}$	

(8.6a)

$FA = (City \times City) + (City \times A)$, the base functor of cons-lists of length at least two; $listcp = wrap + cpr$; $tc : City \times City \rightarrow Real$, neither positive nor symmetric.

```
start (a,b)=([a,b],[a,b]), drop1 (a,([b]*xs,ys))=([a]*xs,[a]*ys), drop2 (a,(xs,[b]*ys))=([a]*xs,[a]*ys),
tour≐([start,drop1 u drop2]).
```

```
cost (xs,ys)=outcost xs+incost ys, outcost [a0,...,an]=tc (a0,a1)+...+tc (an-1,an), incost [a0,...,an]=tc (a1,a0)+...+tc (an,an-1).
```

$$\text{next} \triangleq \text{tail head}, \text{ next2} \triangleq \text{next} \times \text{next}, \text{ head2} \triangleq \text{head} \times \text{head}, R \triangleq \text{cost} \leq \text{cost}^\circ, Q \triangleq R \wedge (\text{next2} \rightarrow \text{next2}^\circ) \wedge (\text{head2} \rightarrow \text{head2}^\circ), P \triangleq T, \\ g_1 \triangleq \text{list}([\text{start}, \text{dropl}]), g_2 \triangleq \text{list}([\text{start}, \text{dropr}]).$$

(8.6b)

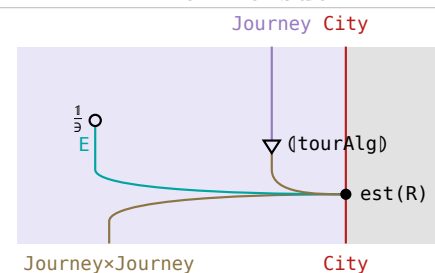
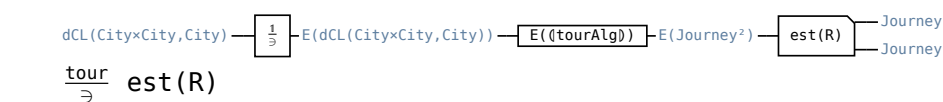
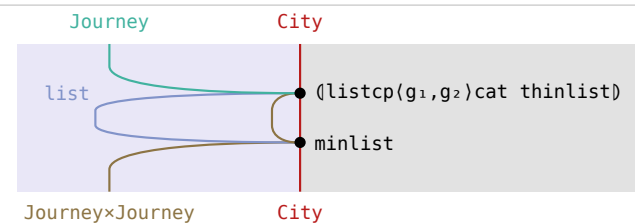
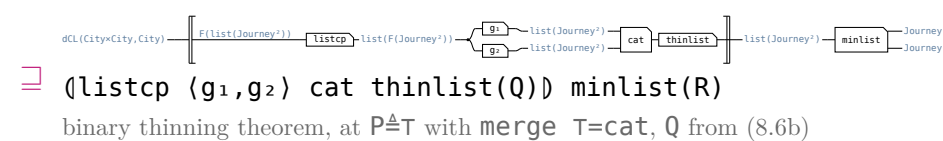
the law	what it says
$(1 \times R) \text{ drop}l \sqsubseteq \text{drop}l \text{ } R$ FALSE $(1 \times R) \text{ drop}r \sqsubseteq \text{drop}r \text{ } R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times Q) \text{ drop}l \sqsubseteq \text{drop}l \text{ } Q$ $(1 \times Q) \text{ drop}r \sqsubseteq \text{drop}r \text{ } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

(8.6c)

$$\begin{aligned}
& (\forall X. \forall Y. X \in Y \implies \text{sortF}(X) \in \text{sortF}(Y)) \wedge \\
& \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \in P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\
& \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \in P([\text{start}, \text{droprFn}]) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \in \text{cp} \\
& \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \in \text{cup} \text{sortP} \wedge \text{sortP} \text{thinlist} \in \text{thin}(\text{Qc}(\text{tc})) \text{sortP} \wedge \text{sortP} \\
& \text{minlist} \in \text{est}(\text{R}) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \in \frac{(\text{tourAlg})}{3} \text{est}(\text{R}) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

circuit

Hinze–Marsden


$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{3}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\langle \langle \text{start}, \text{droplFn} \rangle \cup \langle \text{start}, \text{droprFn} \rangle \rangle)} E(\text{Journey}^?) \xrightarrow{\text{est}(\text{R})} \begin{array}{l} \text{Journey} \\ \text{Journey} \end{array} \\
= & \frac{\langle \langle \text{start}, \text{dropl} \rangle \cup \text{dropr} \rangle}{\ni} \text{est}(\text{R}) \\
& \text{tour} \triangleq \langle \langle \text{start}, \text{dropl} \rangle \cup \text{dropr} \rangle \text{ — } \triangleq \text{tour}
\end{aligned}$$


```

([start wrap,cpr (list(dropl),list(dropr)) cat
thinlist(Q)]) minlist(R)
listcp=wrap+cpr, gi=[list(start),list(dropi)] —  $\triangleq$  tour;
quadratic, two tours added per step

```